

Practical Generative AI for Teaching, Research & Productivity

Powered by Akshara

Target Audience

- Faculty Members
- Researchers
- PhD Scholars
- IT Administrators
- Students (Optional)

Module 1 – Introduction to Generative AI

Topics

- Evolution of AI
- Machine Learning vs Deep Learning
- What is Generative AI?
- Large Language Models (LLMs)
- Transformers (Simplified)
- ChatGPT Overview
- AI Landscape (OpenAI, Gemini, Claude, Perplexity, Grok)

Learning Outcome

Participants will understand how modern AI works and where it is being applied.

Module 2 – Mastering Prompt Engineering

Topics

- What is Prompt Engineering?
- Characteristics of a Good Prompt
- Prompt Frameworks
- Structured Outputs
- Constraints
- Role Prompting
- Chain of Thought Prompting
- Zero-Shot Prompting
- One-Shot Prompting
- Few-Shot Prompting
- Prompt Chaining
- Prompt Libraries

Hands-on

Participants practice writing prompts for:

- Lesson Planning
- Question Paper Generation
- Assignments
- Research
- Coding
- Report Writing

Module 3 – Responsible AI & AI Limitations

Topics

- AI Hallucinations
- Bias in AI
- Detecting AI Generated Content
- Making AI Output Authentic
- Fact Verification
- Ethical AI Usage
- Academic Integrity

Module 4 – Agentic AI + Design Patterns

- Agentic AI fundamentals
- AI agent architecture
- Tool/function calling
- ReAct pattern
- Planning & reasoning
- Reflection / self-correction
- Memory
- RAG agents
- Multi-agent systems
- Agent orchestration
- Supervisor/worker pattern
- Human-in-the-loop
- MCP
- Agent workflows
- State management
- Agent evaluation
- Guardrails
- Production deployment
- Agentic AI design patterns

Module 5 – AI Tools Demonstration

Live demonstrations of:

- ChatGPT
- Gemini
- Claude
- Perplexity AI
- NotebookLM
- Gamma (Presentation Generation)
- Napkin AI (Diagrams)
- Canva AI

Use real academic examples throughout the demos.

Module 6 – Enterprise AI with AINexLayer

From Public AI Tools to Enterprise AI

Demonstration

- Enterprise AI Assistant
- Document Intelligence
- Multi-LLM Support
- Enterprise Search
- Knowledge Base
- RAG (Retrieval-Augmented Generation)
- AI Agents
- Workflow Automation
- Role-Based Access Control
- On-Premise & Cloud Deployment
- Manufacturing, Education, Healthcare, and Enterprise use cases

Show how AINexLayer can support:

- Faculty knowledge assistants
- Research assistants
- Institutional document search
- Academic policy Q&A
- Student support
- Administrative automation

Conclude by showing how the AI Lab infrastructure described in the tender can be used with an enterprise platform like AINexLayer for secure, institution-wide AI adoption.

Q&A

- Open discussion
- Faculty-specific use cases
- Future AI roadmap